Lean Business Canvas · Kasi Grid

Amethyst & Azure · FNB App of the Year Hackathon 2026

PROBLEM

- Unscheduled power and water outages in townships with no published, trusted information (load reduction, cable theft, substation faults; 22 major Johannesburg water outages in two weeks of January 2026).
- Households on medical devices cannot tell if a backup will last.
- Small businesses lose stock and sales and cannot see the cost.

SOLUTION

- Live satellite street with verified outages
- Forecast and typical duration
- Medical battery countdown, alerts and ambulance call
- Care and equipment booking
- Outage-cost planner and restock orders
- Community board; accessibility; four languages

UNIQUE VALUE PROPOSITION

The only app that turns neighbour reports into verified community outage alerts anywhere in South Africa, learns how long they last, and ties them to medical power needs and what an outage costs a business. Free, anonymous, offline.

UNFAIR ADVANTAGE

- Community-generated, device-verified outage history per community that does not exist elsewhere
- Trust in township networks
- A biomedical-engineering view of power dependence
- Works where official data is silent

CUSTOMER SEGMENTS

- Township households (free)
- Medically dependent households and their carers
- Spaza shops and small traders (Pro)
- Suppliers, carers and clinics (partners)
- Municipalities, banks, insurers and NGOs (sponsors)

KEY METRICS

- Verified outages later confirmed (85%)
- Median time to verification (under 20 min)
- Weekly active users and suburbs
- Spoiled stock reduced for spazas (20%)
- Cost per active user (under R5)

CHANNELS

- WhatsApp sharing from the app
- Spaza shop counters (QR code)
- Community groups and ward committees
- Clinics, carers and equipment suppliers
- Installable web app, no app store

COST STRUCTURE

- About R9,800 a month to run a 3,000-user pilot
- About R23,000 one-off (translation review, vetting, POPIA, launch print)
- Moderators, data and outreach are the largest items

REVENUE STREAMS

- Spaza Pro about R49 a month
- Commission on restock orders (4%) and bookings (10%)
- Partner listings about R299 a month
- Municipal dashboard or sponsorship about R3,000 a month

ASSUMPTIONS TO VALIDATE

Hypotheses to validate, not results: spaza willingness to pay, municipal willingness to buy, supplier and carer partnerships, and whether a bank would offer an outage product.

Source code: https://github.com/Unsolvedstar/Kasi-Grid